Study of Static Voltage Stability Region Considering Integration of Renewable Energy Through VSC-HVDC System

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2025 IEEE Power & Energy Society General Meeting (PESGM) | 979-8-3315-0995-8/25/$31.00 ©2025 IEEE | DOI: 10.1109/PESGM52009.2025.11225578

***Abstract*—With the development of large-scale renewable energy sources (RESs) and voltage source converter-based high voltage direct current (VSC-HVDC) technologies, integrating RESs into power systems through VSC-HVDC systems has become increasingly popular. The four-quadrant power control characteristics of VSC converters enable them to function as flexible reactive power sources, which help alleviate voltage stability issues in power systems. However, the implicit functional relationship between the reactive power limits of VSC converters and their output active power, as well as the nonlinear DC power flow between VSC converters in a multi-terminal VSC-HVDC system, make the reactive power boundary of each VSC converter a nonlinear dynamic variable. This adds further complexity to calculating the static voltage stability region boundary (SVSRB). In this paper, an approximate computation model for the SVSR of power systems integrated with large-scale RESs through VSC-HVDC systems is proposed, considering the influences of DC network topology, DC control strategies, control parameters, and other factors. The SVSR is displayed from the perspective of RESs, which helps intuitively evaluate the influence of the DC integration system on the SVSR.**

Keywords—Renewable energy source, VSC-HVDC, static voltage stability region, reactive power limits, VSC converter

I. INTRODUCTION

With the development of power electronic technologies, voltage source converter-based high voltage direct current (VSC-HVDC) transmission has attracted widespread attention for integrating large-scale, remote renewable energy sources (RES). However, the increasing penetration level of renewable energy introduces significant volatility and randomness from large-scale RESs, which has a non-negligible impact on the static voltage stability (SVS) of power systems and can lead to voltage instability [1]. Therefore, evaluating the voltage stability margin (VSM) and preventing the system from voltage collapse have become critical areas of focus for researchers.

A static voltage stability region (SVSR) is a subset of a security region with respect to the active power injection satisfying the voltage stability [2] And the space between the operating point of the system and the SVSR boundary (SVSRB) demonstrates the VSM with respect to a given stress direction [3]. Mathematically, the SVSR is a hyper-surface in parameter space, providing an effective solution to stress direction deviation problems caused by the fluctuating RESs . Existing studies on SVSR are primarily based on pure AC systems [4-5], while research on SVSR considering the integration of RESs with VSC-HVDC systems remains largely unexplored.

The integration of RES into power systems with HVDC systems typically represents them as large power plants, whose capacity is comparable to that of conventional coal-fired power plants. Due to the four-quadrant power control capability of VSC converters, each terminal can continuously and independently control active and reactive power. VSC converters can provide excellent reactive power regulation to the AC system. The work in [6-7] depicted the reactive power boundaries of VSC converters, revealing the complex coupling relationships between the reactive power limits of VSC converters and their output active power. When the DC system is a multi-terminal VSC-HVDC (MTDC) system, the situation becomes more complicated. Due to the connections within the DC network, the active power of each terminal is coupled with others and fluctuates with the RESs. As a result, the reactive power boundary of each VSC converter on the AC system side becomes a nonlinear dynamic variable. The nonlinear dynamic characteristics of the converter reactive power limits further complicate the construction of the static voltage stability region boundary (SVSRB) of the power system. Moreover, to ensure the accuracy of the SVSRB, multiple AC/DC power flow calculations are required to determine the convergence of power flow and to obtain multiple saddle-node bifurcation (SNB) points. This poses a significant challenge of high computational complexity when building the SVSRB using traditional sampling-fitting methods.

In this paper, an approximate computation model for SVSR of the power system integrated with large-scaled RESs by VSC-HVDC system is presented. Firstly, a linearized DC power flow model, considering different DC control strategies,

This project was supported by the National Natural Science Foundation of China (52177097), and the Scientific Research and Innovation Program of the Shanghai Education Commission (2021-01-07-00-07-E00122).

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is developed to obtain the analytical expressions of the active power at the AC system side of the HVDC terminals with respect to variations in RESs. Secondly, an approach searching for dynamic converter reactive power constraints is presented, considering the parameters of the VSC converters and the DC power flow restrictions in the DC network. And then an approximate computation model of SVSR of the power system is developed using the method of PV curves. SVSR of the AC system is depicted from the perspective of RESs for the first time, which helps to evaluate the corresponding Voltage Stability Margin (VSM) of injecting RES power and ensures that the system operates within the SVSR, leveraging the flexible controllability of the VSC-HVDC system.

II Modelling of SVSR for The Power System Integrated with Large-scale Res Through VSC-HVDC

A. Modelling of Linearized DC Power Flow Model for VSC-HVDC System

In a VSC-HVDC system, Generally, centralized DC slack bus control and distributed voltage droop control are the two most commonly used DC voltage control techniques. Therefore, the VSC stations in a DC grid can be categorized into three groups: active power control (APC) converter stations, droop control stations, and the master converter station. To establish a unified model for the linearized DC power flow, the converter stations are first sorted and numbered according to their voltage control strategies. Suppose there are N_{ps} converter stations on the RES side and N_{pcc} stations on the power system side, which include N_p APC stations, N_{vp} droop control stations, and N_v (N_v can be 1 or 0) master stations. The APC stations are further classified into N_{ps} RES-side stations and N_{PN} system-side stations. The APC converter stations are numbered starting from the RES side, beginning with 1, followed by the droop control stations. If a master station exists, it is assigned the last number.So the DC voltage matrix of the VSC-MTDC system $\boldsymbol{U}_d$ can be defined as:

$$\boldsymbol{U}_d=\begin{bmatrix}\boldsymbol{U}_{ps} & \boldsymbol{U}_{pn} & \boldsymbol{U}_{vp} & \boldsymbol{U}_v\end{bmatrix}^T \tag{1}$$

where $\boldsymbol{U}_{ps}$ and $\boldsymbol{U}_{pn}$ represent the voltage vectors of the APC converter stations connecting to the RES and power system, respectively. $\boldsymbol{U}_{vp}$ and U_v are the voltage vectors of DC droop buses and the slack bus, respectively.

The topology of the DC grid is defined by a node-branch correlation matrix $\boldsymbol{A}$, and

$$\boldsymbol{A}=\begin{bmatrix}a_{ij}\end{bmatrix}_{(N_{ps}+N_{pcc})\times H} \tag{2}$$

where H is the number of DC lines. Matrix element a_{ij} represents the connection and power direction of the line j and the converter station i.

$$a_{ij}=\begin{cases}1, \text{branch } j \text{ starts from station } i\\ -1, \text{branch } j \text{ ends to station } i\\ 0, \text{station } i \text{ is not the end of branch } j\end{cases} \tag{3}$$

Suppose that R_1, R_2,... and R_H represent the resistances of the DC branches. Then the conductance matrix of the DC network can be obtained:

$$\boldsymbol{Y}_d=\boldsymbol{A}\boldsymbol{Y}_s\boldsymbol{A}^T \tag{4}$$

where $\boldsymbol{Y}_s$ is a diagonal matrix.

When only resistance properties are considered in the DC network, $\boldsymbol{Y}_d$ can be expressed in the form of the self-conductance and mutual conductance of DC nodes according to their control techniques. So.

$$\boldsymbol{Y}_d=\begin{bmatrix}\boldsymbol{G}_{ps\text{-}ps} & \boldsymbol{G}_{ps\text{-}pn} & \boldsymbol{G}_{ps\text{-}vp} & \boldsymbol{G}_{ps\text{-}v}\\ \boldsymbol{G}_{pn\text{-}ps} & \boldsymbol{G}_{pn\text{-}pn} & \boldsymbol{G}_{pn\text{-}vp} & \boldsymbol{G}_{pn\text{-}v}\\ \boldsymbol{G}_{vp\text{-}ps} & \boldsymbol{G}_{vp\text{-}pn} & \boldsymbol{G}_{vp\text{-}vp} & \boldsymbol{G}_{vp\text{-}v}\\ \boldsymbol{G}_{v\text{-}ps} & \mathrm{G}_{v\text{-}pn} & \boldsymbol{G}_{v\text{-}vp} & \mathrm{G}_{v\text{-}v}\end{bmatrix} \tag{5}$$

where $\boldsymbol{G}_{ps\text{-}ps}$, $\boldsymbol{G}_{pn\text{-}pn}$, $\boldsymbol{G}_{vp\text{-}vp}$, and $G_{v\text{-}v}$ are the self-conductance matrices, and $\boldsymbol{G}_{ps\text{-}pn}$, $\boldsymbol{G}_{ps\text{-}vp}$, $\boldsymbol{G}_{ps\text{-}v}$, $\boldsymbol{G}_{pn\text{-}ps}$, $\boldsymbol{G}_{vp\text{-}pn}$, $\boldsymbol{G}_{vp\text{-}v}$, $\boldsymbol{G}_{v\text{-}ps}$, $\boldsymbol{G}_{v\text{-}pn}$, and $\boldsymbol{G}_{v\text{-}vp}$ are the mutual conductance matrices The subscripts indicate the different types of DC nodes.

Thus, the DC power flow equations can be expressed as follows:

$$\boldsymbol{P}_d=\boldsymbol{U}_d'\boldsymbol{Y}_d\boldsymbol{U}_d \tag{6}$$

where $\boldsymbol{U}_d'=\mathrm{diag}\left\{\boldsymbol{U}_{ps},\boldsymbol{U}_{pn},\boldsymbol{U}_{vp},U_v\right\}$

From (6), nonlinear items in the DC power flow equations can be identified. And also the unknown variables according to the types of DC nodes. Considering that variations of DC voltage are limited in a small range during the static operation, the voltage at DC node i satisfies $U_{di}\approx 1.0\text{p.u.}$. Thus,

$$U_{di}*U_{dj}\approx U_{dj} \tag{7}$$

where U_{di} and U_{dj} are the voltage magnitudes of the DC bus i and j, respectively.

Since the DC voltage at the master station is almost constant and kept at a reference value, the equations in (9) can be written as:

$$\boldsymbol{P}_d=\boldsymbol{U}_d''\boldsymbol{Y}_d\boldsymbol{U}_d \tag{8}$$

where $\boldsymbol{U}_d''=\mathrm{diag}\left\{\boldsymbol{1},\boldsymbol{1},\boldsymbol{1},U_v\right\}$, $\boldsymbol{P}_d=\begin{bmatrix}\boldsymbol{P}_{ps} & \boldsymbol{P}_{pn} & \boldsymbol{P}_{vp} & P_v\end{bmatrix}^T$

After removing the equation of master station from (8), it can be obtained that:

$$\begin{bmatrix}\boldsymbol{P}_{ps}\\ \boldsymbol{P}_{pn}\\ \boldsymbol{P}_{vp}\end{bmatrix}=\begin{bmatrix}\boldsymbol{P}_{ps,set}\\ \boldsymbol{P}_{pn,set}\\ \boldsymbol{P}_{vp,set}+\boldsymbol{k}_p(\boldsymbol{U}_{dref}-\boldsymbol{U}_{vp})\end{bmatrix} \tag{9}$$

where $\boldsymbol{P}_{ps,set}$ and $\boldsymbol{P}_{pn,set}$ are vectors of active power references of the APC stations connecting to the RES and the power grid, respectively. $\boldsymbol{P}_{vp,set}$, $\boldsymbol{k}_p$, and $\boldsymbol{U}_{dref}$ are vectors of active power reference, droop gains, and voltage set-points of the DC droop stations.

Combining (8) and (9), the voltage vector of the DC system can be written as:

$$\begin{bmatrix}\boldsymbol{U}_{ps}\\ \boldsymbol{U}_{pn}\\ \boldsymbol{U}_{vp}\end{bmatrix}=\begin{bmatrix}\boldsymbol{G}_{ps\text{-}ps} & \boldsymbol{G}_{ps\text{-}pn} & \boldsymbol{G}_{ps\text{-}vp}\\ \boldsymbol{G}_{pn\text{-}ps} & \boldsymbol{G}_{pn\text{-}pn} & \boldsymbol{G}_{pn\text{-}vp}\\ \boldsymbol{G}_{vp\text{-}ps} & \boldsymbol{G}_{vp\text{-}pn} & \boldsymbol{G}_{vp\text{-}vp}+\boldsymbol{k}_p\end{bmatrix}^{-1}\begin{bmatrix}\boldsymbol{P}_{ps,set}-U_v\boldsymbol{G}_{v\text{-}ps}\\ \boldsymbol{P}_{pn,set}-U_v\boldsymbol{G}_{v\text{-}pn}\\ \boldsymbol{P}_{vp,set}+\boldsymbol{k}_p\boldsymbol{U}_{dref}-U_v\boldsymbol{G}_{v\text{-}vp}\end{bmatrix} \tag{10}$$

Observing that $\boldsymbol{P}_{vp}$ is the only vector to be calculated, it can be written as:

$$\boldsymbol{P}_{vp}=\boldsymbol{P}_{vp,set}+\boldsymbol{k}_p\boldsymbol{U}_{dref}-\boldsymbol{k}_p\begin{bmatrix}\boldsymbol{C}_7 & \boldsymbol{C}_8 & \boldsymbol{C}_9\end{bmatrix}\begin{bmatrix}\boldsymbol{P}_{ps,set}-U_v\boldsymbol{G}_{v\text{-}ps}\\ \boldsymbol{P}_{pn.set}-U_v\boldsymbol{G}_{v\text{-}pn}\\ \boldsymbol{P}_{vp,set}+\boldsymbol{k}_p\boldsymbol{U}_{dref}-U_v\boldsymbol{G}_{v\text{-}vp}\end{bmatrix} \tag{11}$$

where $\boldsymbol{C}_7$, $\boldsymbol{C}_8$, and $\boldsymbol{C}_9$ are matrix elements that can be calculated by:

$$\begin{bmatrix}\boldsymbol{C}_1 & \boldsymbol{C}_2 & \boldsymbol{C}_3\\ \boldsymbol{C}_4 & \boldsymbol{C}_5 & \boldsymbol{C}_6\\ \boldsymbol{C}_7 & \mathrm{C}_8 & \boldsymbol{C}_9\end{bmatrix}=\begin{bmatrix}\boldsymbol{G}_{ps\text{-}ps} & \boldsymbol{G}_{ps\text{-}pn} & \boldsymbol{G}_{ps\text{-}vp}\\ \boldsymbol{G}_{pN\text{-}pS} & \boldsymbol{G}_{pn\text{-}pn} & \boldsymbol{G}_{pn\text{-}vp}\\ \boldsymbol{G}_{vp\text{-}ps} & \boldsymbol{G}_{vp\text{-}pn} & \boldsymbol{G}_{vp\text{-}vp}+\boldsymbol{k}_p\end{bmatrix}^{-1} \tag{12}$$

Finally, combination (13) and (11), P_v can be calculated by:

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$$P_{\mathrm{v}}=U_{\mathrm{v}}\begin{bmatrix}\boldsymbol{G}_{\text{v-ps}}\\ \boldsymbol{G}_{\text{v-pn}}\\ \boldsymbol{G}_{\text{v-vp}}\end{bmatrix}^{\mathrm{T}}\begin{bmatrix}\boldsymbol{C}_1 & \boldsymbol{C}_2 & \boldsymbol{C}_3\\ \boldsymbol{C}_4 & \boldsymbol{C}_5 & \boldsymbol{C}_6\\ \boldsymbol{C}_7 & \mathrm{C}_8 & \boldsymbol{C}_9\end{bmatrix}\begin{bmatrix}\boldsymbol{P}_{\text{ps,set}}\text{-}U_{\mathrm{v}}\boldsymbol{G}_{\text{v-ps}}\\ \boldsymbol{P}_{\text{pn,set}}\text{-}U_{\mathrm{v}}\boldsymbol{G}_{\text{v-pn}}\\ \boldsymbol{P}_{\text{vp,set}}+\boldsymbol{k}_{\mathrm{p}}\boldsymbol{U}_{\text{dref}}\text{-}U_{\mathrm{v}}\boldsymbol{G}_{\text{v-vp}}\end{bmatrix}+U_{\mathrm{v}}{}^2G_{\text{v-v}} \quad (13)$$

Because the VSC stations on the side of power grid are the only stations impacting SVSR directly, the active power interchanges between the VSC-HVDC system and the AC power system is identified and denoted as $\boldsymbol{P}_{\mathrm{ac}}$. So, there is:

$$\begin{cases}\boldsymbol{P}_{\mathrm{ac}}=\begin{bmatrix}\boldsymbol{P}_{\mathrm{pn}}\\ P_{\mathrm{v}}\end{bmatrix}=\begin{bmatrix}\boldsymbol{0}\\ \boldsymbol{C}_{\mathrm{p}}\end{bmatrix}\boldsymbol{P}_{\text{ps,set}}+U_{\mathrm{v}}\begin{bmatrix}\boldsymbol{C}_{\text{v1}} & \boldsymbol{0}\\ \boldsymbol{C}_{\text{v2}} & \boldsymbol{C}_{\text{v3}}\end{bmatrix}\begin{bmatrix}\boldsymbol{P}_{\text{pn,set}}\\ U_{\mathrm{v}}\end{bmatrix}\\ \boldsymbol{C}_{\mathrm{p}}=U_{\mathrm{v}}\left(\boldsymbol{C}_1+\boldsymbol{C}_4\right)\boldsymbol{G}_{\text{v-ps}}\\ \boldsymbol{C}_{\text{v1}}=\boldsymbol{I}_{\text{pn}}U_{\mathrm{v}}{}^{-1}\qquad \boldsymbol{C}_{\text{v2}}=\left(\boldsymbol{C}_2+\boldsymbol{C}_5\right)\boldsymbol{G}_{\text{v-pn}}\\ \boldsymbol{C}_{\text{v3}}=\boldsymbol{G}_{\text{v-v}}+\boldsymbol{G}_{\text{v-ps}}\left(\boldsymbol{C}_1\boldsymbol{G}_{\text{v-ps}}+\boldsymbol{C}_2\boldsymbol{G}_{\text{v-pn}}\right)+\boldsymbol{G}_{\text{v-pn}}\left(\boldsymbol{C}_5\boldsymbol{G}_{\text{v-pn}}+\boldsymbol{C}_4\boldsymbol{G}_{\text{v-ps}}\right)\end{cases} \quad (14)$$

where $\boldsymbol{I}_{\mathrm{pn}}$ is a $N_{\mathrm{pn}}\times 1$ column vector with all the elements being 1.

If there is no slack bus in the DC system ($N_{\mathrm{v}}=0$), in the DC system with a droop control scheme. $\boldsymbol{P}_{\mathrm{AC}}$ can be written as

$$\boldsymbol{P}_{\mathrm{ac}}=\begin{bmatrix}\boldsymbol{P}_{\mathrm{pn}}\\ \boldsymbol{P}_{\mathrm{vp}}\end{bmatrix}=\begin{bmatrix}\boldsymbol{0}\\ \text{-}\boldsymbol{k}_{\mathrm{p}}\boldsymbol{C}_7\end{bmatrix}\left[\boldsymbol{P}_{\text{ps,set}}\right]+\begin{bmatrix}\boldsymbol{I}_{\mathrm{vp}} & \text{-}\boldsymbol{k}_{\mathrm{p}}\boldsymbol{C}_8\\ \boldsymbol{0} & \boldsymbol{k}_{\mathrm{p}}(\boldsymbol{I}_{\mathrm{vp}}\text{-}\boldsymbol{k}_{\mathrm{p}}\boldsymbol{C}_9)\\ \boldsymbol{0} & \boldsymbol{I}_{\text{dvp}}\text{-}\boldsymbol{k}_{\mathrm{p}}\boldsymbol{C}_9\end{bmatrix}^{\mathrm{T}}\begin{bmatrix}\boldsymbol{P}_{\text{pn,set}}\\ \boldsymbol{U}_{\text{dref}}\\ \boldsymbol{P}_{\text{vp,set}}\end{bmatrix} \quad (15)$$

where $\boldsymbol{I}_{\mathrm{vp}}$ is a $N_{\mathrm{vp}}\times 1$ column vector with all the elements being 1.

According to the characteristics of VSC-HVDC system, $\boldsymbol{Y}_{\mathrm{d}}$, $\boldsymbol{P}_{\text{vp,set}}$, $\boldsymbol{k}_{\mathrm{p}}$, $\boldsymbol{U}_{\text{dref}}$, $\boldsymbol{P}_{\text{pn,set}}$, and U_{v} are all known. Comparing (14) with (15), the power delivered to the AC network can be expressed as a function of the active power of RESs and the DC voltages. Thus, a genetic linearized DC power flow model can be obtained.

$$\boldsymbol{P}_{\mathrm{ac}}=\boldsymbol{k}_{\mathrm{n}}\boldsymbol{P}_{\mathrm{ps}}+\boldsymbol{C}_{\mathrm{n}} \quad (16)$$

B. Modelling of dynamic reactive power boundary of VSC converter at PCCS

Fig.1 illustrates a general representation of a VSC station connecting to an AC power system. In Fig.1, X_{eq} is the equivalent reactance of the converter transformer and the phase reactor, and X_{s} and R_{s} are the equivalent reactance and resistance of the AC system. k_{T} is the ratio of the transformer. m_{U} is the voltage modulation ratio, and i_{DC} and U_{DC} are the DC current and DC voltage, respectively. U_{c} is the voltage of the inverter output, U_{s} is the equivalent voltage source of the AC system, and U_{p} is the voltage at the PCC. P_{pcc} and Q_{pcc} are the active and reactive power delivered from the VSC converter to the receiving grid, respectively. P_{c} and Q_{c} are the active and reactive power at the inverter output. Here, U_{c} is set to be the reference .So, $U_{\mathrm{c}}=U_{\mathrm{c}}\angle 0^{\circ}$, $U_{\mathrm{p}}=U_{\mathrm{p}}\angle\delta_1$, $U_{\mathrm{s}}=U_{\mathrm{s}}\angle\delta_2$.

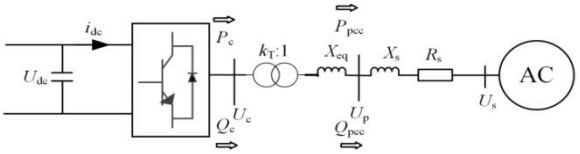


Fig.1. Steady-state model of a VSC station

Because the modulation ratio of voltage in the VSC converter $m_{\mathrm{U}}\in[0.85, 1]$, the relationship between the output active power and the reactive power of a VSC can be concluded:

$$\begin{cases}\left(P_{\mathrm{c}}-\dfrac{3(m_{\mathrm{U}}U_{\mathrm{dc}})^2}{8k_{\mathrm{T}}{}^2Z}\cos\varphi\right)^2+\left(Q_{\mathrm{c}}-\dfrac{3(m_{\mathrm{U}}U_{\mathrm{dc}})^2}{8k_{\mathrm{T}}{}^2Z}\sin\varphi\right)^2=\left(\dfrac{\sqrt{3}m_{\mathrm{U}}U_{\mathrm{dc}}U_{\mathrm{s}}}{2\sqrt{2}k_{\mathrm{T}}Z}\right)^2\\ 0.85\le m_{\mathrm{U}}=\dfrac{2\sqrt{2}U_{\mathrm{c}}}{\sqrt{3}U_{\mathrm{dc}}}\le 1\qquad Z\angle\varphi=R_{\mathrm{s}}+j(X_{\mathrm{s}}+X_{\mathrm{eq}})\end{cases} \quad (17)$$

There are some other constraints that must be considered in the static operations of a VSC converter. Firstly, the output active and reactive power of a converter station should not exceed its rated capacity S. So,

$$P_{\mathrm{c}}{}^2+Q_{\mathrm{c}}{}^2\le S^2 \quad (18)$$

Secondly, the operating current of the converter should stay below the upper limits of the electrical devices. So,

$$\begin{cases}P_{\mathrm{c}}{}^2+Q_{\mathrm{c}}{}^2\le\left(U_{\mathrm{c}}I_{\mathrm{xmax}}\right)^2\\ P_{\mathrm{c}}\le U_{\mathrm{dc}}I_{\mathrm{dcmax}}\\ I_{\mathrm{b}}\le I_{\mathrm{bmax}}\end{cases} \quad (19)$$

where I_{xmax} and I_{dcmax} represent the maximal current on the AC and DC side of the converter, respectively. And I_{b} and I_{bmax} represent the operating current and its maximal value in the bridge arm. And

$$\begin{cases}I_{\mathrm{b}}=\dfrac{I_{\mathrm{dc}}}{3}+\dfrac{I_{\mathrm{x}}}{2}\\ U_{\mathrm{dc}}I_{\mathrm{dc}}=\sqrt{3}U_{\mathrm{c}}I_{\mathrm{x}}\cos\zeta\end{cases} \quad (20)$$

where I_{x} is the operating current of the converter, ζ represents the power factor angle.

Combining (20) and (21), it can be concluded that:

$$\begin{cases}P_{\mathrm{c}}{}^2+Q_{\mathrm{c}}{}^2\le\left(ZJ+\dfrac{U_{\mathrm{s}}{}^2+\sqrt{(2ZJ+U_{\mathrm{s}})^2-4Z^2(P_{\mathrm{c}}{}^2+Q_{\mathrm{c}}{}^2)}}{2}\right)I_{\mathrm{xmax}}^2\\ P_{\mathrm{c}}\le U_{\mathrm{dc}}I_{\mathrm{dcmax}}\\ P_{\mathrm{c}}{}^2+Q_{\mathrm{c}}{}^2\le\left(\dfrac{2\sqrt{3g}U_{\mathrm{dc}}I_{\mathrm{bmax}}}{2\sqrt{g}\cos\zeta+\sqrt{6}U_{\mathrm{dc}}}\right)\\ g=ZJ+\dfrac{(U_{\mathrm{s}})^2+\sqrt{(2ZJ+U_{\mathrm{s}})^2-4Z^2(P_{\mathrm{c}}{}^2+Q_{\mathrm{c}}{}^2)}}{2}\\ J=P_{\mathrm{c}}\cos\varphi+Q_{\mathrm{c}}\sin\varphi\end{cases} \quad (21)$$

In a typical MMC-based VSC system, the arbitrary phase contains two arms, each consisting of N_0 identical sub-modules (SMs). During operation, , there are differences in turn-on time and conduction currents between SMs in the same arm.The SM voltage ripples should be kept within a certain range, which is expressed in terms of energy.

Take A-phase as an example, the fluctuating energy in an arm can be expressed as:

$$\Delta W=\int_{-\arccos(\frac{1}{m_{\mathrm{I}}})-\zeta-\pi}^{\arccos(\frac{1}{m_{\mathrm{I}}})-\zeta-\pi}|\,p_{\mathrm{pa}}\,|\,d(\omega t)=\frac{S}{3m_{\mathrm{U}}\omega}\left[1-(\frac{m_{\mathrm{U}}\cos\zeta}{2})^2\right]^{\frac{3}{2}} \quad (22)$$

where p_{pa} represents the instantaneous active power of A-phase bridge arm m_{I} represents the modulation ratio of current.

Taking the voltage fluctuation threshold to be ε, then

$$\Delta W\le 2N_0C_0U_0{}^2\varepsilon \quad (23)$$

Substitute (22) in (23) to obtain:

$$\frac{\sqrt{P_{\mathrm{c}}{}^2+Q_{\mathrm{c}}{}^2}}{3}\cdot\frac{\left[1-\left(\dfrac{m_{\mathrm{U}}\cos\zeta}{2}\right)^2\right]^{\frac{3}{2}}}{m_{\mathrm{U}}N_0C_0U_0{}^2\omega}\le\varepsilon \quad (24)$$

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where C_0 is the capacitance of a SM, U_0 is the voltage of the sub-modules, and ω is the angular frequency of the system.

Finally, according to the power exchanges between the VSC converter and the AC system, in order to guarantee the existence of a real root for U_c, the following conditions must be me

$$\begin{cases} c \ge 0 \qquad U_s^{\,2} + 2a + \sqrt{c} \ge 0 \\ c = \left(U_s^{\,2} + 2a\right)^2 - 4\left(a^2 + b^2\right) \\ a = P_c R_s + Q_c(X_{eq} + X_s) \qquad b = P_c(X_{eq} + X_s) - Q_c R_s \end{cases} \tag{25}$$

In general, the reactive power constraints of a converter can be depicted as a two-dimensional space composed of an upper and a lower boundary, which can be traced by a set of output active and reactive power that satisfies (32).

$$\left(P_{pcc}, Q_{pcc}\right) \in \left\{ \begin{array}{l} P_{pcc} = P_c \\ Q_{pcc} = P_c - \dfrac{P_c^2 + Q_c^2}{(U_c / k_T)^2} X_{eq} \end{array} \middle| \begin{array}{l} s.t.(17),(18), \\ (21),(24),(25) \end{array} \right\} \tag{26}$$

Obviously, the limits of Q_{pcc} change with fluctuating P_{pcc} due to RESs. Since it is not easy to obtain the analytical expression of the boundaries, a tracing method is introduced to develop the computation model of the dynamic reactive power boundaries. The procedure is as follows::

Step 1: Generate a finite set of P_c+jQ_c representing the output active and reactive power of the VSC converter while considering the capacity restriction.

Step 2: Substitute combinations of P_c and Q_c into (26) successively to calculate and screen out the sets of P_{pcc} and Q_{pcc} to form the operation region of the VSC converter.

Step 3: Find the 2-D boundary of the operation region and fit the upper and lower boundary curves using polynomial approximation. The upper and lower boundary curves are defined as Q_{pcc}^{+} and Q_{pcc}^{-}.So;

$$\begin{cases} Q_{pcc}^{\ +} = f_u(P_{pcc}) \\ Q_{pcc}^{\ -} = f_l(P_{pcc}) \end{cases} \tag{27}$$

where f_u and f_l represent the fitting functions of the upper and lower boundary, respectively.

In a MTDC system, the DC power flow is determined by the voltages of the converters and the resistors of the DC lines. When the converter losses are ignored, the active power injected into the VSC converter on the DC side will be transmitted to the power system entirely.While (27) presents the active power relations between the VSC converters, the upper and lower reactive power boundaries of the VSC converters connecting to the PCCs can be obtained by substituting (16) into (27):

$$\begin{cases} Q_{pcc,i}^{\ +} = f_{u,i}\left(iloc\left[\boldsymbol{k}_n \boldsymbol{P}_{ps} + \boldsymbol{C}_n\right]_i\right) \\ Q_{pcc,i}^{\ -} = f_{l,i}\left(iloc\left[\boldsymbol{k}_n \boldsymbol{P}_{ps} + \boldsymbol{C}_n\right]_i\right) \end{cases} \tag{28}$$

where $i \in 1,2,\ldots,N_{pcc}$. $f_{u,i}$ and $f_{l,i}$ represent the fitting functions of the upper and lower reactive power boundaries at the i-th PCC. $iloc[.]_i$ represents a mathematical symbol for taking the i-th element from the row vector.

C. Approximate Model of SVSR

MIIF is a parameter for estimating the voltage interactions between converters in the same AC system. In this paper, MIIF is used to decouple the system from the AC side at the PCCs.

$$MIIF_{ji} = \frac{Z_{ij}}{Z_{ii}} \tag{29}$$

where Z_{ii} is the equivalent self-impedance of node i .Z_{ij} is the equivalent mutual impedance between nodes i and j.

the power injected into the PCC i can be expressed as

$$P_i^{'} + \mathrm{j}Q_i^{'} = P_i + jQ_i + MIIF_{ji}(P_j + jQ_j) \tag{30}$$

Then the voltage at the PCC i can be obtained:

$$\begin{aligned} U_i^{\ 2} = {} & \frac{E_i^{\ 2}}{2} + P_i^{'} R_{ii} + Q_i^{'} \\ & \pm\sqrt{\frac{E_i^{\ 4}}{4} + E_i^{\ 2} P_i^{'} R_{ii} + E_i^{\ 2} Q_i^{'} X_{ii} - \left(P_i^{'} X_{ii} - Q_i^{'} R_{ii}\right)^2} \end{aligned} \tag{31}$$

With the resistance ignored, Based on PV curve threshold the SVSRB of the PCC$_i$ can be written as:

$$f_i = P_i^{'} - \frac{\sqrt{E_i^{\ 4} + 4E_i^{\ 2} X_{ii} Q_i^{'}}}{2X_{ii}} \tag{32}$$

On the basis of the above analysis, the SVSRB of the power system considering the variations of RESs can be written as:

$$\begin{aligned} f_i = {} & iloc\left[\boldsymbol{k}_n \boldsymbol{P}_{ps} + \boldsymbol{c}_n\right]_i + \sum_{j=1 j\neq 1}^{N_{pcc}} \frac{X_{ij}}{X_{ii}} \cdot iloc\left[\boldsymbol{k}_n \boldsymbol{P}_{ps} + \boldsymbol{c}_n\right]_i \\ & - \frac{\sqrt{E_i^{\ 4} + 4E_i^{\ 2} X_{ii}\left(Q_{pcc,i}^{\ +} + \sum_{j=1 j\neq 1}^{N_{pcc}} \frac{X_{ij}}{X_{ii}} Q_{pcc,j}^{\ +}\right)}}{2X_{ii}} \end{aligned} \tag{33}$$

where $i \in 1,2,\ldots,N_{pcc}$.

So the whole SVSR can be represented by the space of $\boldsymbol{P}_{ps}$.

$$\Omega = \left\{ (P_{ps,1}, \cdots, P_{ps,N_{ps}}) \in R^{N_{ps}} \middle| \begin{array}{l} f_i(P_{ps,1}, \cdots, P_{ps,N_{ps}}) \le 0 \\ P_{ps,i} \le S_{ps,i} \end{array} \right\} \tag{34}$$

where $i \in 1,2,\cdots,N_{pcc}$. R^{Nps} describes the power in RESs of each control command variable. $S_{ps,i}$ represents the capacity of the i-th converter station on the RESs side

III Case Studies

The proposed method will be validated with a four-terminal HVDC system, as illustrated in Fig.2 and the detailed data are presented in Tables I.

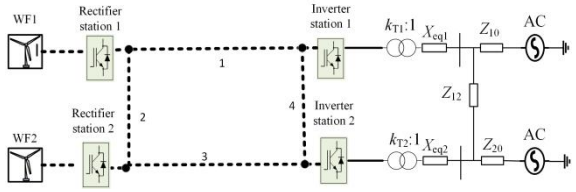


Fig.2. Simplified connection diagram of RESs and power system through a four-terminal HVDC system

TABLE I
PARAMETERS OF CONVERTER STATIONS IN SYSTEM

Parameters	U_{dc} (kV)	k_T	I_{dcmax} (p.u.)	I_{vmax} (p.u)	I_{bmax} (p.u)	X_{eq} (p.u)	C_0 (F)	N_0
value	50	1.1	1.1	1.0	3.125	0.2	0.007	100
Parameters	U_0 (kV)	ε	S (MVA)	K_p	U_{dref} (kV)	U_v (kV)	$P_{ps,set1}$ (MW)	$P_{ps,set2}$ (MW)
value	5	0.08	600	200	500	500	600	500
Parameters	$P_{vp,set1}$ (MW)	$P_{vp,set2}$ (MW)	$P_{pn,set}$ (MW)	Z_{10} (H)	Z_{20} (H)	Z_{12} (H)	E_1 (kV)	E_2 (kV)
value	600	600	500	0.3H	0.36	0.36	250	250

A. Verification of the Linearized DC Power Flow Model

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The effectiveness and accuracy of the derived general expressions for injecting active power at PCCs are first demonstrated and analyzed. As a comparison, the DC power flow is calculated using the Newton-Raphson method. To quantitatively evaluate the precision, the relative standard deviation (RSD) is introduced to present the errors, where the baseline capacity S_B=400MW.

$$RSD = \frac{\text{Calculated value} - \text{Reference value}}{\text{Reference value}} \times 100\% \quad (35)$$

The results and errors of the two methods of two typical HVDC control strategies are presented in Fig.3. The computing times are presented in Table II. The proposed linearized DC power flow calculation method has advantages in computation speed while keeping the RSDs within 0.5%.

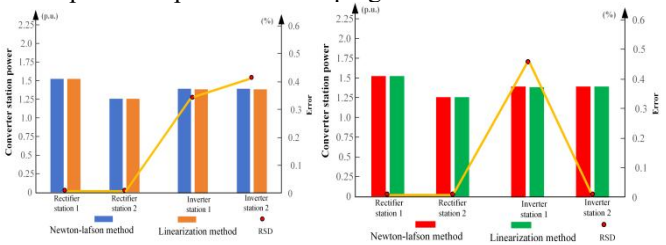


Fig.3. Comparison of DC power flow results . (a) Calculation results of the drooping control strategy. (a) Calculation results of the master -slave control strategy .

TABLE II

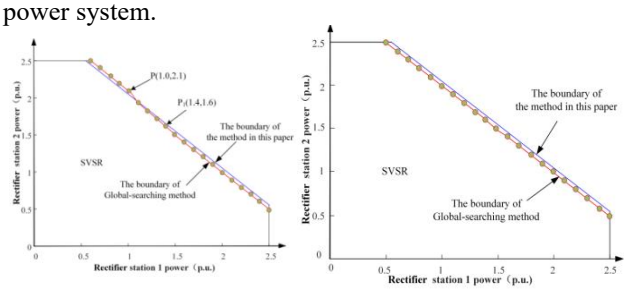
COMPARISONS OF COMPUTATIONAL TIMES

method	Proposed model (s)	Newton–Raphson method (s)
the drooping control strategy	0.082	0.291
the master -slave control strategy	0.083	0.283

B.Analysis of SVSR

In order to validate the proposed computation model in (34), a global searching method is introduced to construct SVSR by searching the SVSRB with increasing output active power of RESs from 0.1 p.u. to the rated power. The voltage collapse points of the power system are calculated one by one using the method presented in [8]. Both master-slave control schemes and droop control schemes are considered here. Fig.4(a) and (b) illustrate the SVSR regions obtained by the two methods for the two types of DC voltage control schemes, respectively. As shown, the calculated results obtained by the proposed approximate computation model fit the global-searching results quite well, both in the master-slave control scheme and the droop control scheme. Table III presents the excellent computation performance of the proposed method.

Moreover, it can be observed from Fig.4 that: 1) The proposed model demonstrates the SVSR of the power system from the RES side, not the PCC side. It is a two-dimensional space constructed by the active power injections of the VSC converters on the side of RESs. Thus, it can help evaluate the maximum renewable power penetration of the power system with the given PCCs and display the corresponding VSM with respect to the varying RESs. 2) Since there are two VSC stations on the RES side, the SVSR is a two-dimensional space composed of the intersection area of two voltage instability curves of the PCCs and two straight lines representing the rated capacities of the VSC stations on the RES side.. 3) Optimizing the distribution of active power between APC stations on the RES side may be helpful to improve the maximum renewable power penetration of the power system.



(a) (b)

Fig.4. Results of SVSR of the power system considering the influences of DC control strategies . (a) Comparisons of SVSR with DC system under drooping control scheme.(b) Comparisons of SVSR with DC system under master -slave control scheme

TABLE III

COMPARISONS OF SVSR CALCULATION TIME

method	Method of this paper(s)	Global-searching method(s)
droop control scheme	1.23	51.3
master-slave control scheme	1.24	52.6

III. CONCLUSION

In this paper, an approach for developing an approximate computation model of SVSR considering the integration of large-scale RESs through a VSC-HVDC system is proposed. The characteristics of the VSC-HVDC system, including the DC network, DC control strategy, and control parameters, are considered in the modeling of SVSR for the first time. The high precision and fast calculation speed of the method are also demonstrated.

The proposed approach displays the SVSR from the perspective of RESs instead of the PCCs in the power system. This allows for an intuitive evaluation of the corresponding VSM when injecting RES power and ensures that the operation of RESs remains within the SVSR, leveraging the flexible control capabilities of the VSC-HVDC system to their fullest extent.

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